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Review

Drug development for refractory epilepsy: The past 25 years and beyond

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ABSTRACT

Despite the exponential growth of approved antiepileptic drugs (AEDs) over the past 25 years, epilepsy remains uncontrolled in approximately a third of patients. This article summarises the clinical trials and properties of the AEDs developed over this period, and reviews the pre-clinical and clinical development paradigms of modern AEDs. We discuss possible reasons for the apparent failure to develop more efficacious compounds. We also review the current regulatory frameworks for drug approval in the United States and Europe, and the changes on the horizon. Encouragingly, better elucidation of the pathophysiological mechanisms underpinning pharmacoresistance and the epilepsies by recent research has enabled a revised approach to the development of more promising therapies. A new era of pharmacological treatment for epilepsy appears imminent. Future research in pharmacotherapy for drug-resistant epilepsy will be advanced through concerted effort between scientists, clinicians, and the industry.

1. Introduction

The mainstay of treatment for epilepsy is pharmacological therapy for seizure control. Driven by the limited efficacy of the established antiepileptic drugs (which generally include carbamazepine, ethosuximide, phenobarbital, phenytoin and valproic acid), antiepileptic drug (AED) development has exploded in the past 25 years (Fig. 1). But despite the release of a new agent almost annually over this period, epilepsy remains uncontrolled in one third of patients [1]. Uncontrolled epilepsy is associated with poorer quality of life, increased physical and psychological comorbidities, and increased risk of sudden unexplained death in epilepsy, placing substantial burden on the individuals, carers, and society [2,3].

This article reviews the pre-clinical and clinical development paradigms of modern AEDs, and summarises the clinical trials and properties of the newer AEDs approved for the treatment of drug-resistant epilepsy over the past 25 years. We discuss possible reasons for the lack of fundamental improvement in treatment efficacy, and provide an overview of promising research directions in pharmacotherapy for drug-resistant epilepsy.

2. Preclinical drug development paradigms

Three main approaches have been employed to identify compounds with potential anti-seizure activity: random screening of synthesized chemicals, structural variations of known AEDs, and rational drug development through selective targeting of seizure-inducing mechanisms [4]. All approaches rely on the preclinical use of animal models to establish safety and efficacy of the investigational compounds prior to human trials. The maximal electric shock (MES) and the subcutaneous pentylenetetrazol (scPTZ) rodent models have traditionally been the first-line models for the discovery of new AEDs. In the MES model, seizures are induced by bilateral trans-auricular or corneal electrical stimulation. The model tends to identify drugs that block generalized tonic-clonic seizures [4,5]. However, several newer AEDs (such as vigabatrin and levetiracetam) with demonstrated efficacy for focal seizures in human epilepsy are ineffective in the MES model. In the scPTZ model, clonic seizures are evoked through subcutaneous administration of convulsant doses of PTZ, which acts as a $GABA_A$ receptor antagonist [6]. The scPTZ model tends to identify drugs

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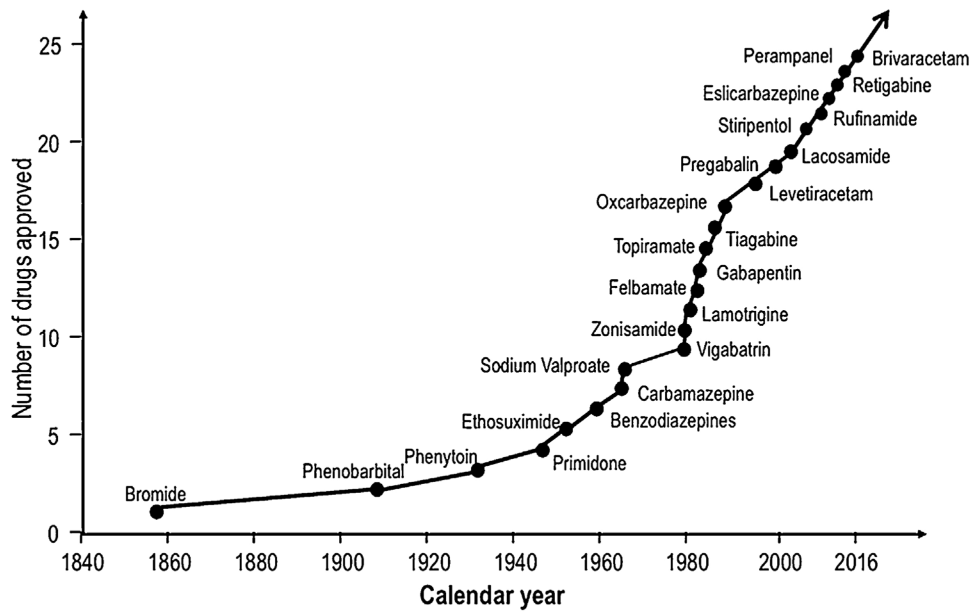

Fig. 1. Chronological development of antiepileptic drugs.

Adapted from Kwan [43]

that block generalized non-convulsive (myoclonic, absence) seizures.

The MES and scPTZ models have proven useful for detecting anti-seizure effects of drugs in healthy rodents, and have provided insight into the pharmacokinetic and pharmacodynamic interactions of potential AEDs [4]. However, both models can produce false negative results, the notable example being levetiracetam. Importantly, these acute seizure models are not designed to identify compounds which act to inhibit spontaneous seizures or drug-resistant epilepsy [7]. These models may also fail to detect potentially efficacious compounds that act via mechanisms not present in these models [6]. Since chemicals not effective in these models are excluded from further development, this is of significant concern. The diversity of human epilepsy syndromes poses a need for animal models that more fully capitulate the clinical scenarios.

To address this need, other animal models have recently been developed to supplement the MES and scPTZ tests [4]. These include the 6 Hz psychomotor seizure model, phenytoin-resistant kindled rat, lamotrigine kindled-rat, post-status epilepticus models of temporal lobe epilepsy, and the methylazozymethanol (MAM) model [4]. In particular, the 6 Hz test has been readily adopted in the National Institutes of Health AED discovery program. This model administers low-frequency and long-duration electrical stimulation via corneal electrodes to induce focal seizures [8]. It successfully detects the efficacy of levetiracetam, which is ineffective in the MES and scPTZ models [4]. The kindling model aims to mimic the development of focal epilepsy through repeated applications of subthreshold electrical stimulations to specific brain regions (usually the amygdala or hippocampus) to induce increasingly severe seizure behaviour. It has been noted, however, that AEDs which exhibit high efficacy in the kindling model do not necessarily have higher clinical efficacy in patients with drug-resistant focal seizures.

3. Limitations of animal models in AED development

The MES and scPTZ preclinical models have proven useful for identifying various new AEDs, but they have not resulted in the development of AEDs with higher efficacy than established agents in drug-resistant patients. This is likely due to the intrinsic limitations of these models. First, seizure definitions used in experimental animal models are limited to specific convulsive phenotypes and minimum durations, and fail to account for the short non-convulsive seizures often observed in human focal epilepsies. Second, these preclinical models were originally validated by the older AEDs, which may explain the failure to identify novel AEDs with better efficacy and tolerability [4]. This contributes to the redundancy of "me-too" drugs. Third, the determination of drug efficacy is dependent upon the ability of an investigational drug to inhibit provoked seizures, whereas the seizures in human epilepsy are characteristically unprovoked and originate from a molecular substrate. However, it has been observed that the pharmacology of the provoked seizures in the kindling model and the spontaneous seizures in the post-status epilepticus model is similar, suggesting that the model's brain alterations may be more important than the method of seizure induction [9].

4. Clinical drug development paradigms

4.1. Adjunctive therapy

Due to ethical reasons, investigational compounds are initially tested as adjunctive treatment in patients with uncontrolled epilepsy. Similar paradigms have been adopted by the US Food and Drug Administration (FDA) and the European Medicines Agency (EMA) for clinical AED development. Generally speaking, a new molecular agent is required to undergo at least two independent studies before regulatory submission can be accepted for approval. Regulatory trials are randomized, double-blind and placebo-controlled, and usually evaluate a number of fixed doses [10]. A Phase III trial typically includes a baseline period, a titration period and a maintenance period. The titration period involves increasing the dose of the drug up to the maximal tolerated dose, or a pre-defined fixed dose, which is maintained for usually 12–16 weeks. Although there is no regulatory guidance regarding forced or flexible dose titration, both the FDA and EMA prefer pre-defined target doses, covering the purportedly minimal effective dose to the maximal tolerated dose [11]. Phase III programs often include an open label extension phase to assess long-term safety and seizure outcomes.

There are minor differences in the primary efficacy outcome measure accepted by the agencies: the EMA prefers responder rate (percentage of patients with at least 50% reduction in seizure frequency during the treatment phase compared with the baseline), while the FDA requires outcomes to be measured as median percent seizure reduction.

Table 1 summarizes the phase III clinical trials that evaluated investigational AEDs in the past 25 years. The pharmacological properties of the AEDs approved in this period are provided in Table 2 [12,13].

4.2. Monotherapy

Following the approval of an AED as adjunctive therapy, its further development for monotherapy use is under different regulatory frameworks in the US and Europe. The FDA requires that trials show superiority of the investigational drug over a comparator agent. However, randomizing patients to placebo has ethical concerns. Several new study designs were developed to comply with this regulation, but were eventually abandoned for ethical reasons. These included the once widely implemented "pseudo-placebo" design, in which a small dose of a single agent is given to patients in one arm [14].

In contrast to the FDA, the EMA accepts direct comparison of a new drug with an existing drug. In this paradigm, demonstration of equivalence or non-inferiority is adequate for approval. Several drugs have gained monotherapy approval using the non-inferiority design, including levetiracetam [15] and zonisamide [16], which both used a flexible-dosing design and controlled-release carbamazepine as the comparator. The main concern with non-inferiority designs, and the reason why the FDA requires a superiority outcome, is their unproved assay sensitivity. This is based on the argument that equivalent efficacy in the two trial arms could have been similarly achieved by a placebo.

4.3. The historical-control design

In the US, monotherapy efficacy assessment in drug-resistant patients had traditionally employed either "withdrawal-to-monotherapy" or "withdrawal-to-placebo." These approaches have raised ethical concerns and may not adequately reflect intended use, and have thus attracted widespread criticism [17]. Recently, the historical-control approach has attempted to overcome the ethical difficulties of monotherapy vs. placebo trials [18]. In this approach, drug-resistant patients are withdrawn to the study drug in monotherapy, and compared to a historic control group modelled from previous conversion-to-monotherapy study data. This design has been adopted for evaluating monotherapy indication of eslicarbazepine acetate, lacosamide, lamotrigine extended release, levetiracetam extended release and pregabalin [19–23]. However, this design has inherent concerns relating to the inconsistency between study cohorts and time-dependent population changes. Additionally, the lack of a parallel control group prevents effective blinding of treatment (some trials used blinded doses). For these reasons, the EMA regards historical-control studies as complementary rather than conclusive for the assessment of monotherapy indication [24].

4.4. Should separate monotherapy indication be re-examined?

To expedite monotherapy approval, Mintzer et al. recently proposed re-examination of the policy of separate approvals for AEDs as monotherapy and adjunctive therapy [25]. It is noted that AEDs are the only neurotherapeutics with separate indications for monotherapy and adjunctive use, which creates complications to the approval process. They suggest that regulatory restrictions that prevent or delay monotherapy approval for valuable new drugs is harmful to patients (for example, levetiracetam is not granted monotherapy indication in the US). They propose that AEDs should be approved for the treatment of specific seizure types, with approval for monotherapy and adjunctive use granted simultaneously. Whether the FDA can be persuaded remains to be seen. However, sequential licensing has recently been accepted by the EMA [26]. The AED under examination would be approved for adjunctive and second-intention monotherapy use, and then granted unrestricted monotherapy license after successfully adhering to a series of safety checkpoints. No monotherapy AED approval has been granted under these new pathways yet.

5. Future research directions

Despite the introduction of over 15 new compounds in the past 25 years, the overall proportion of patients with refractory epilepsy has remained largely unchanged. A shift from symptomatic seizure control to targeting underlying biological mechanisms is needed. The transition towards the development of disease-modifying treatments would be accelerated by a better understanding of the heterogeneous nature of epilepsy and the multiple complex factors that contribute to the abnormal neural discharges. Table 3 lists selected new compounds currently under early clinical investigation.

5.1. Promising approaches for pharmacological therapy

There is a need to elucidate the pathology underpinning pharmacoresistance to develop novel therapies. Several hypotheses have been formulated. These include the target hypothesis, the multidrug transporter hypothesis and the network hypothesis [6,27,28]. The target hypothesis suggests that alteration of molecular drug targets results in reduced response to treatment. This hypothesis is based on studies of AED effects on hippocampal voltage-gated sodium channels [29,30]. The multidrug transporter hypothesis postulates that an overexpression of efflux transporters at the blood–brain barrier restricts penetration of AEDs and reduces drug concentrations to subtherapeutic levels. Bankstahl et al. addressed this hypothesis by determining the efficacy of six AEDs in wildtype mice and mice deficient in P-glycoprotein (Pgp), one of the best studied efflux transporters, in a kainate-induced status epilepticus model of mesial temporal lobe epilepsy [31]. No significant differences in anti-seizure drug efficacies were observed between wildtype and Pgp-deficient mice. While this finding does not invalidate the multi-drug transporter hypothesis, it suggests that Pgp may not be functionally relevant in the mechanisms underlying drug resistant seizures. The co-administration of transporter inhibitors with AEDs is a promising therapeutic avenue to overcome drug resistance, but further research into other efflux transporters is needed. The network hypothesis suggests that neurodegeneration occurs in drug-resistant brains, and postulates that drugs with neuroprotective actions may be beneficial. Stemming from this hypothesis, immunosuppressive treatments and inhibitors of inflammation have been considered as potential therapeutic options. They include inhibitors of cytokine synthesis, neuroimmunophilins, corticosteroids and interleukin-1 receptor antagonists [32].

Agents with disease-modifying effects are of emergent interest in epilepsy treatment research. Although evidence for anti-

Table 1
Summary of double-blind, randomized phase III trials of antiepileptic drugs for treatment of refractory partial (focal) onset seizures (+/− other seizure types).

Drug	Seizure type	Study design (number of trials performed)	Total population (number of patients in active arm(s))	Comparator	Primary outcome measure	Reference(s)
Vigabatrin	Partial onset	Adjunctive, cross-over (n = 3)	n = 23 (n = 23); n = 31 (n = 31); n = 97 (n = 97)	Placebo	Decrease in total number of seizures; percent change in mean weekly seizure frequency; monthly seizure rate, calculated for each of the four-week study periods	[44–46]
		Adjunctive (n = 1)	n = 111 (n = 58)	Placebo	Decrease in median monthly seizure frequency	[47]
Lamotrigine	Partial onset	Immediate release (n = 1)	n = 156 (n = 76)	Low-dose VPA	Proportion of patients meeting exit criteria	[48]
	Idiopathic generalized, mixed seizure types	Adjunctive, cross over (n = 1)	n = 26 (n = 26)	Placebo	Percent reduction in seizure rate for individual seizure types	[49]
	Primary generalized tonic-clonic seizures	Adjunctive (n = 1)	n = 121 (n = 58)	Placebo	Median percent change from the baseline phase in the average monthly primary generalized tonic-clonic seizures seizure frequency	[50]
	Partial onset	Historical pseudo-placebo Extended release (n = 1)	n = NA (n = 174)	Historical pseudo placebo	Proportion of patients meeting exit criteria	[21]
Oxcarbazepine	Partial onset	Pre-surgical (n = 1)	n = 102 (n = 51)	Placebo	Time to meet one of exit criteria	[51]
		Conversion-to monotherapy (n = 2)	n = 87 (n = 41); n = 143 (n = 96)	2400 vs. 300 mg/day	Percent of patients meeting one of four exit criteria; time to meet one of four exit criteria	[52,53]
Felbamate	Partial onset	Adjunctive, cross-over (n = 1)	n = 30 (n = 30)	Placebo	Number of seizures experienced by the patient during each of the analysis periods	[54]
		Adjunctive, pre-surgical (n = 1)	n = 64 (n = 30)	Placebo	Time to fourth seizure	[55]
		Conversion-to-monotherapy (n = 2)	n = 44 (n = 22); n = 111 (n = 56)	Low dose VPA	Number of patients in each treatment group who met escape criteria; number of patients in each group who met escape criteria	[56,57]
		Pre-surgical (n = 2)	n = 52 (n = 25); n = 40 (n = 21)	Placebo	Average daily seizure frequency during treatment; average frequency of all seizure types among study completers	[58,59]
Gabapentin	Partial onset	Conversion-to-monotherapy, dose-controlled (n = 1)	n = 274 (n = 91)	2400 vs 1200 vs 600 mg/day	Time to exit from double-blind phase	[60]
		Pre-surgical, dose-controlled (n = 1)	n = 82 (n = 40)	3600 vs 300 mg/day	Time to exit from study	[61]
		Adjunctive (n = 1)	n = 209 (n = 127)	Placebo	Response ratio, calculated according to the RRatio formula	[62]
Topiramate	Partial onset	Adjunctive (n = 3)	n = 181 (n = 136); n = 47 (n = 23); n = 41 (n = 20)	Placebo	Percent reduction in average monthly seizure rate; (1) median percent seizure reduction (2) ≥50% reduction; proportion of patients with ≥50% seizure reduction	[63–65]
	Partial onset	Conversion-to-monotherapy (n = 1)	n = 46 (n = 23)	1000 vs 100 mg/day	Fulfilment of exit criteria	[66]
	Generalized tonic-clonic seizures	Adjunctive (n = 1)	n = 88 (n = 39)	Placebo	Percentage reduction in PGTC seizure rate during double-blind phase	[67]
	Partial onset	Adjunctive (n = 1)	n = 249 (124)	Placebo	Median percent reduction in weekly partial onset seizure frequency.	[68]
Tiagabine	Partial onset	Pre-surgical (n = 1)	n = 11(n = 7)	Placebo	Median reduction in 4-week complex partial seizure rates from baseline to the combined titration and fixed-dose periods for each treatment group	[69]
		Low vs high dose (n = 1)	n = 198 (n = 96)	36 vs 6 mg/day	Median reduction in 4-week complex partial seizure rates from baseline to the combined titration and fixed-dose periods for each treatment group	[69]
		Adjunctive, crossover (n = 1)	n = 94,open screening phase (n = 46, double-blind cross-over phase)	Placebo	Four-week seizure frequency in the double-blind crossover phase of the study	[70]
		Adjunctive (n = 2)	n = 154 (n = 77); n = 297 (n = 206)	Placebo	Proportion of responders (patients achieving a reduction of 50% or more in the 4-weekly seizure rate) calculated for all partial seizures and for each seizure subtype; median change in 4-week seizure frequency	[71,72]

Levetiracetam	Partial onset	Adjunctive (n = 3)	n = 294 (n = 199); n = 324 (n = 212); n = 56 (n = 28)	Placebo	Mean number of partial seizures per week; weekly partial seizure frequency; logarithmically transformed weekly frequency of partial seizures over the 16-week, double-blind treatment phase	[73–75]
	Partial onset	Conversion-to-monotherapy (n = 1)	n = 286 (n = 181)	Placebo	Percentage of patients who completed monotherapy phase relative to number of patients randomized to receive medication	[76]
	Generalized tonic-clonic seizures	Adjunctive (n = 1)	n = 164 (n = 80)	Placebo	Percentage reduction from the combined baseline period in GTC seizure frequency per week	[77]
	Partial onset	Conversion-to-monotherapy, Historical pseudo-placebo, extended release (n = 1)	n = NA (n = 171)	Historical pseudo-placebo	Cumulative exit rate at day 112 due to exit criteria compared with historical control	[19]
Zonisamide[a]	Partial onset	Adjunctive (n = 2)	n = 139 (n = 71); n = 152 (n = 78)	Placebo	Median percentage reduction from baseline in seizure frequency	[78,79]
Stiripentol[a]	Partial onset	Adjunctive, enrichment and withdrawal (n = 1)	n = 67 (n = 17)	Placebo	Number of patients meeting the escape criteria during the double-blind period	[80]
Pregabalin	Partial onset	Adjunctive (n = 5)	n = 453 (n = 353); n = 287 (n = 191); n = 313 (n = 225); n = 341 (n = 268); n = 178 (n = 119)	Placebo	Reduction in seizure frequency as measured by RRatio; reduction in seizure frequency during treatment as compared to baseline; change in seizure frequency from baseline; reduction in seizure frequency as measured by RRatio; reduction in seizure frequency as measured by RRatio	[81–85]
		Historical pseudo-placebo (n = 1)	n = NA (n = 161)	Historical pseudo-placebo	Proportion of the pregabalin 600 mg/d group meeting ≥ 1 of the predefined seizure-related exit criteria	[20]
Rufinamide[a]	Partial onset	Adjunctive (n = 2)	n = 313 (n = 156); n = 357 (n = 176)	Placebo	Percent change in partial seizure frequency	[86,87]
Lacosamide	Partial-onset	Adjunctive (n = 3)	n = 497 (n = 421); n = 485 (n = 318); n = 405 (n = 301)	Placebo	(1) change in seizure frequency per 28 days from baseline to maintenance (2) reduction of at least 50% in seizure frequency from baseline to maintenance	[10,88,89]
		Conversion-to-monotherapy, historical control (n = 1)	n = NA (n = 425)	Historical pseudo-placebo	Kaplan–Meier-predicted percentage of patients on 400 mg/day in the full analysis set meeting ≥ 1 predefined seizure-related exit criterion by day 112, compared with the historical-control threshold	[23]
		Adjunctive (n = 1)	n = 548 (n = 364)	Placebo	Change in partial onset frequency per 28 days from Baseline to the Maintenance period	[90]
Eslicarbazepine acetate	Partial onset	Adjunctive (n = 3)	n = 402 (n = 300); n = 253 (n = 165); n = 395 (n = 295)	Placebo	Seizure frequency per 4 weeks; seizure frequency as recorded by patient diaries; seizure frequency as recorded by patient diaries	[91–93]
		Conversion-to-monotherapy, historical pseudo-placebo (n = 2)	n = NA (n = 172); n = NA (n = 193)	Historical pseudo-placebo	Kaplan–Meier-estimated 112-day exit rate with a threshold value calculated from the historical controls; proportion of patients meeting predefined exit criteria	[22,94]
Retigabine/ezogabine[a]	Partial onset	Adjunctive (n = 4)	n = 399 (n = 303); n = 538 (n = 359); n = 305 (n = 153); n = 75 (n = 50)	Placebo	Percentage change from baseline in monthly seizure frequency and compared across treatment arms; change in 28-day seizure frequency; (1) percent change in 28-day total partial-seizure frequency from baseline to 18-week double-blind period (2) responder rate, defined as the proportion of patients experiencing a ≥50% reduction in 28-day total partial-seizure frequency from baseline to maintenance phase; proportion of responders, defined as patients with ≥50% reduction in 28-day total partial-onset	[95–98]

Table 1 (*Continued*)

Drug	Seizure type	Study design (number of trials performed)	Total population (number of patients in active arm(s))	Comparator	Primary outcome measure	Reference(s)
					seizure frequency, from the baseline phase to the maintenance phase	
Perampanel[a]	Partial onset	Adjunctive (n = 4)	n = 387 (n = 266); n = 706 (n = 521); n = 386 (n = 250); n = 162 (n = 81)	Placebo	The responder rate (percentage of patients who experienced a 50% reduction in seizure frequency in the maintenance period relative to baseline); median percent change in seizure frequency; (1) percent change in 28-day total partial-seizure frequency from baseline to 18-week double-blind period (2) responder rate, defined as the proportion of patients experiencing a ≥50% reduction in 28-day total partial-seizure frequency from baseline to maintenance phase; Percent change in PGTC seizure frequency	[99–102]
Brivaracetem[a]	Partial onset	Adjunctive (n = 4)	n = 396 (n = 298); n = 480 (n = 359); n = 398 (n = 298); n = 768 (n = 501)	Placebo	Focal seizure frequency per week; median percent reduction in baseline-adjusted seizure frequency; focal seizure frequency per week; (1) percent reduction in 28-day adjusted partial onset seizure frequency (2) ≥50% responder rate based on percent reduction in seizure frequency from baseline to the treatment period	[103–106]

[a] Approved for adjunctive use only; RRatio = (T − B)/(T + B), where T and B are the seizure frequencies during treatment and during baseline.

epileptogenic effect of approved AEDs (including levetiracetam, phenobarbital, valproate and topiramate) is present in the post-status epilepticus animal models, this property has not been demonstrated in humans. These models have also been employed to screen for disease-modifying effects of non-AED compounds. Lorsartan is an example of such a compound, and has been shown to inhibit epileptogenic mechanisms *in vivo* [33]. Rapamycin and derivative molecules (everolimus, temsirolimus, deforolimus, ridaforolimusm) inhibit the mTOR signalling system and represent another potential class of epileptogenic compounds. They have been explored in animal models of tuberous sclerosis complex and some models of acquired epilepsy, and display promising results in human trials of tuberous sclerosis complex [34].

The blood–brain barrier is also under investigation as a possible target. Agents such as doxycline and minocycline that affect blood–brain barrier permeability are potential anti-epileptogenic candidates. Other compounds that have been of emerging interest are anti-apoptotic agents (such as corticotropin releasing hormone, and some AEDs), antioxidants (lipoic acid, adenosine, melatonin, vitamins C and E) and activators of neurotrophic receptors (fibroblast growth factor 2, brain derived neurotrophic factor, neuropeptide Y, inhibitors of TrkB kinase, erythropoietin) [35].

Lastly, there has been a resurgence of interest in the proposed anti-seizure properties of cannabidiol, derived from the cannabis plant [36,37]. Interestingly, cannabidiol's anti-seizure properties are not mediated by an interaction with cannabinoid receptors, despite its structural similarity to tetrahydrocannabinol [38]. Pre-clinical evaluations have confirmed these properties, and cannabidiol is now being subject to various clinical trials. In addition to its potential as an anti-seizure agent, cannabidiol has been shown to possess neuroprotective and anti-inflammatory effects. Promising results were found in a recently completed open-label study investigating the use of cannabidiol oral solution as an adjunctive therapy for paediatric and adult patients with drug-resistant epilepsy [37]. A number of randomized controlled trials are underway to elucidate the efficacy and safety profile of cannabidiol.

5.2. Precision medicine

Individual response to pharmacological treatment is influenced by the genetic profile. Pharmacogenomics is a new addition to epilepsy research, offering the potential to personalise treatment for individuals with epilepsy. The EMA defines pharmacogenomics as the "study of the variability of individual genes relevant to disease susceptibility as well as drug response at cellular, tissue, individual or population level" [39]. Taking this approach further, precision medicine aims to stratify individuals into subpopulations based on differences in disease susceptibility, prognosis and treatment response. Precision medicine thus promises to maximise outcomes and minimise unnecessary adverse drug reactions and expenditure, providing clinical and socioeconomic benefits for both the individual patients and the community. A battery of genetic tests is currently used for screening epilepsy susceptibility genes and to characterize gene variant profiles at the family and population level. It is proposed that by identifying the causative mutation for an individual epilepsy case, targeted therapeutic choices can be made. Under this paradigm, clinical practice will no longer be directed by the suboptimal "trial and error" methods.

The repurposing of existing drugs not currently indicated for epilepsy has the potential to overcome drug resistance.

Table 2
Properties of modern antiepileptic drugs (adapted from Kwan et al. [126] and Brodie and Kwan [12]).

Drug	Year of approval	Primary mechanism(s) of action	Indications (types of seizures/syndromes)	Absorption (bioavailability %)	Protein binding (% bound)	Elimination half-life (h)	Metabolism and routes of elimination
Vigabatrin	1989	GABAergic	Partial onset, West syndrome	Slow (60–80)	0	5–7	Not metabolized, renal excretion
Lamotrigine	1990	Sodium-channel blockade	Partial, generalised, Lennox-Gastaut	Rapid (95–100)	55	22–36	Glucuronidation
Oxcarbazepine	1990	Sodium-channel blockade	Partial, GTCS	Rapid (95–100)	40	8–10	Hepatic conversion to active moiety
Felbamate	1993	Multiple	Partial onset, Lennox-Gastaut	Slow (95–100)	22–36	13–23	Hepatic metabolism, renal excretion
Gabapentin	1993	Neuronal calcium-channel binding	Partial onset	Slow (60)	0	6–9	Not metabolised, renal excretion
Topiramate	1995	Multiple	Partial onset, GTCS, myoclonic, Lennox-Gastaut	Slow (80)	9–17	20–24	Hepatic metabolism, renal excretion
Tiagabine	1996	GABAergic	Partial onset	Rapid (95–100)	96	5–9	Hepatic metabolism
Levetiracetam	2000	Binds to SV2A receptors	Partial onset, GTCS, myoclonic	Rapid (95–100)	<10	7–8	Non-hepatic hydrolysis, Renal excretion
Zonisamide	2000	Multiple	Partial onset, GTCS, myoclonic	Rapid (95–100)	40–60	50–68	Hepatic metabolism, renal excretion
Stiripentol	2002	GABAergic	Dravet Syndrome	Rapid (>70)	99	8.5–23.5	Demethylation, glucuronidation, renal excretion.
Pregabalin	2004	Neuronal calcium-channel binding	Partial onset	Rapid (90–100)	0	6	Renal excretion
Rufinamide	2004	Sodium-channel blockade	Lennox-Gastaut, Partial onset	Slow (>85)	34	6–10	Hepatic metabolism
Lacosamide	2008	Slow inactivation of sodium channel/interacts with CRMP-2	Partial onset	Rapid (95–100)	<15	13	Hepatic metabolism
Eslicarbazepine acetate	2009	Sodium-channel blockade	Partial onset, GTCS	Rapid (90)	40	13–20	Glucuronidation, renal excretion
Retigabine/ ezogabine	2011	Activation of low-threshold potassium channels	Partial onset	Rapid (60)	80	8	Glucuronidation
Perampanel	2012	Non-competitive AMPA-receptor antagonist	Partial onset	Rapid (100)	95	105	Glucoronidation, feces, urine
Brivaracetam	2016	Binds to SV2A receptors	Partial onset	Rapid (100)	<20	7–8	Renal excretion

AMPA, alpha-amino-3-hydroxy-5-methyl-4-isoxazole propionic acid; CRMP-2, collapsin response mediator protein-2; GABA, gamma-aminobutyric acid; GTCS, generalized tonic-clonic seizures; SV2A, synaptic vesicle 2A.

Table 3
Selected compounds currently under clinical investigation.

Developmental approach	Name of compound	Mechanism of action	References
Mechanisms of action similar to those of marketed AEDs	Ganaxolone	GABA receptor modulator	[107–109]
	Allopregnanolone (SAGE-547)	$GABA_A$ receptor modulator	[110]
	Selurampanel (BGG492)	Competitive antagonist for AMPA and kainate receptors	[111]
	ICA-105665	Selective opener of neuronal Kv7 potassium channels	[112]
	YKP3089 (cenobamate)	Selective blocker for the inactivated state of the sodium channel, facilitates presynaptic GABA release	[113]
Novel mechanisms of action	Beprodone	Agonist of the melatonin type 3 receptor	[113]
	Huperzine A	Inhibitor of AChE receptor	[113]
Repurposed compounds which were initially developed for treatment of other diseases	Everolimus	Selective inhibitor of mTOR pathway	[114]
	Fenfluramine	Serotonin reuptake inhibitor	[115–117]
	Nalutozan	Nonazapirone 5-$HT1_A$ partial agonist	[118]
	Pitolisant	Histamine 3 receptor antagonist	[119]
	Quinidine	Partial antagonist of KCNT1	[120]
	Valnoctamide	$GABA_A$ receptor agonist	[121]
	Verapamil	Inhibitor of P-glycoprotein	[122,123]
Unknown mechanisms of action	JNJ-26489112	Multiple, unknown	[124]

Personalized genomic medicine was successfully implemented in the diagnosis and treatment in a recent case of drug-resistant infantile-onset epilepsy [40]. A novel mutation was found in the GRIN2A gene, coding for the NMDA receptor (which has a pivotal role in neuronal communication). Following administration of the Alzheimer disease drug memantine, which reduces chronic overstimulation of NMDAR, a reduction in seizure frequency was observed. Another important example of drug repositioning in precision medicine is the use of quinidine. Case reports show that this cardiac anti-arrhythmia drug is effective in epilepsy syndromes with an underlying gain-of-function mutation in the KCNT1 gene [41]. Quinidine inhibits the KCNT1 sodium-activated potassium channel, encoded by the KCNT1 gene. It is hypothesized that quinidine may counteract the effect of the mutation of the channel.

6. Conclusion

While the lack of substantial progress of treatment for drug-resistant epilepsy over the past two decades has been attributed to deficiencies in the preclinical and clinical stages of development, there is a sense that a new era of treatment for drug-resistant therapy is imminent. A better understanding of the underlying physiological mechanisms leading to epilepsy has permitted the move towards developing novel target-driven therapeutic strategies [42]. This may allow optimization of the translation of preclinical findings to clinical research through thorough validation of novel drug-targets prior to extensive drug discovery work. The multifactorial mechanisms underpinning drug resistance will continue to demand collective effort from basic science and clinical research.

Conflict of interest statement

PK has received research grants from the National Health and Medical Research Council of Australia, the Australian Research Council, RMH Neuroscience Foundation, the US National Institutes of Health, Bill & Melinda Gates Foundation, Hong Kong Research Grants Council, Innovation and Technology Fund, Health and Health Services Research Fund, and Health and Medical Research Fund. He/his institution also received speaker or consultancy fees and/or research grants from Eisai, GlaxoSmithKline, Johnson & Johnson, Pfizer, and UCB Pharma.

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